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Analysing Reddit Communities and Their Impact on Dogecoin Price Dynamics

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Abstract

This thesis investigates whether behavioral shifts in Reddit community users can help predict Dogecoin price dynamics. By combining network analysis to find vital nodes with efficient change point detection methods, this analysis finds that including the behavioral shifts in the predictive models increases their performance, particularly for return, and especially around higher uncertainty periods. The results highlight the relevance of studying online communities for financial market analysis.

1 Introduction

The rise of retail investment and the dissemination of social media have changed the dynamics of major financial markets. Big platforms like X (former Twitter) and Reddit have become established sites of discussion, exchange of ideas and opinions, investment strategies and market rumours, allowing for collective biases and sentiment to proliferate faster than ever and threaten the assumptions of traditional finance. The meme asset phenomenon, which includes cryptocurrencies such as Dogecoin, is a perfect example of how online communities can drive considerable fluctuations in prices and trading activity that are not explained or detected by traditional financial methodologies. Understanding the behavior and impact of these online communities in some assets is, therefore, a topic of very high relevance.

This thesis focuses on analyzing how the structure and functioning of a Reddit community and its most influential users can impact metrics such as return, trading volume, and volatility of Dogecoin. The analysis does not simply monitor the overall activity of the network, it goes further by identifying the points in time where major behavior changes occur, focusing on the patterns of interactions of the most influential users - vital nodes. The main research question is to understand to what extent the identified shifts in collective behavior, more than just volume of online interactions, can predict Dogecoin price dynamics.

A comprehensive data-driven methodology was built to answer this question. The study begins by collecting 8 months of posts and comments data from r/dogecoin, the subreddit specific for Dogecoin discussion. Vital nodes - the network's most influential nodes - are then identified according to their network activity and posting consistency and regularity. After this, the Pruned Exact Linear Time (PELT) algorithm, which is exact and very efficient, is used to detect change points in these users' interactions, i.e. moments in time where an abrupt change in activity occurred. The change points, alongside a measure

of overall network activity, are then used to define rolling window regression models that predict Dogecoin's return, trading volume, and volatility.

The structure of the thesis starts with a review of relevant previous research and theoretical background that inspired and motivated this work. It then includes a chapter on data processing, explaining how the data was collected and prepared for the analysis, and then proceeds to the methodology chapter, where the whole framework is described and explained in detail, before presenting the results and the respective insights and conclusions. There is also a robustness analysis chapter in the end, as well as an appendix with some relevant formulas.

2 Theoretical background

The rise of social media platforms has transformed the propagation of financial information from contemporaneous markets. Traditional financial theories are based in the efficient market hypothesis (Sewell [2011]), which assumes that the market fully reflects the information available, i.e. all participants have access to the entire relevant information. However, the digital age introduced new information channels, such as social media platforms, that challenge this traditional market theory by creating potential behavioral biases and public sentiment that influence financial asset prices.

Recent evidence shows that social media sentiment can have a significant impact on market volatility and the trading patterns of the investors. Studies have pointed that Twitter-based sentiment analysis can predict unexpected returns in big stock indexes, depending on the market conditions (Kim and Kauffman [2024]). Social media influences investors differently when compared traditional news channels, generating phenomenons like investment cascades that disturb the dynamics of the market (Bikhchandani and Sharma [2001]). These phenomenons are especially visible in the context of retail investment, where

platforms such as Reddit have become hubs of financial discussion.

There has been extensive research in the behavioral finance field around the impact of psychological biases in investment decisions, and social media magnifies this impact. Research has shown that investors who use social media for investment information are more likely to invest in cryptocurrencies than those who don't (Kim and Fan [2025]). This digitization of financial advice brought challenges to traditional investment models, and the effect of social media in some financial assets can no longer be ignored.

Reddit, in particular, has grown as a particularly influential platform for financial discussions, which was arguably potentiated by its structure of posts, with topic-specific subreddits that allow for focused discussion and create a network where collective sentiment and biases can quickly rise and propagate.

The r/wallstreetbets phenomenon, especially during the GameStop (GME) short squeeze in 2021, showed that Reddit communities can have a clear impact on price movements of relevant financial assets. Previous research used over 1.6 million r/wallstreetbets posts to show that financial recommendations on Reddit can compete with the biggest investment banks, even outperforming them in certain situations (Buz and de Melo [2022]).

There have also been studies suggesting that the effectiveness of sentiment analysis, particularly on meme stocks, decreases around the big frenzies and major market events, with lots of content posted to grab attention that disturbs the accuracy of the sentiment analysis.

Bhatt et al. [2023] demonstrates that incorporating Twitter sentiment analysis into Bitcoin price prediction models significantly improves forecasting accuracy, achieving up to 90% accuracy when simultaneously using social media sentiment and traditional market indicators. The influence of social media goes beyond individual sentiment, also extending to collective action, as evidenced by Pelster and Gonzalez [2016], which provide a theoretical basis for the impact of coordinated social media activity on market dynamics, showing

that social media interactions amplify decision-making biases, with users who have a large amount of followers showing a stronger presence of these biases.

To complement the inefficacy of sentiment analysis in a frenzy context, and leveraging the hierarchic post and comment structure of the subreddits, network analysis methods were introduced (Trimborn and Zwetsloot [2024]). These network analysis methods allow for a better modeling of the structure of social relationships and interactions, which can then be used to predict major shifts in user behavior through change point identification. The application of such methods to the cryptocurrency market, specifically to meme coins such as Dogecoin, comes up as an interesting and unexplored field.

Change point detection has been a commonly used process to detect shifts in statistical properties of time series. The fundamental problem, formulated in the general changepoint detection framework, seeks to identify time points in which the statistical properties of a series change abruptly.

Early developments in change point detection included the binary segmentation method, which identifies the best change point in a segment and then iteratively repeats this process for any subsegment resulting from the identified change points. A change point at position τ is detected when

$$\mathcal{C}(y_{1:\tau}) + \mathcal{C}(y_{(\tau+1):n}) + \beta < \mathcal{C}(y_{1:n}) \tag{1}$$

While this approach is computationally efficient, with an $\mathcal{O}(n \log n)$ complexity, it finds only an approximate solution to the optimization problem.

The optimal partitioning method addresses the limitations of binary segmentation by using dynamic programming to find an optimal solution to the minimization of the penalized segment cost function. This framework was first proposed by Jackson et al. [2004] and finds the optimal set of change points by solving the following minimization problem sequentially:

$$F(t) = \min_{0 \leq \tau < t} \left[F(\tau) + C\left(y_{(\tau+1):t}\right) + \beta \right] \tag{2}$$

However, this method's complexity $\mathcal{O}(n^2)$ hampers its applicability to large datasets.

The Pruned Exact Linear Time (PELT) algorithm, introduced by Killick et al. [2012], combines the exactness of optimal partitioning with improved computational efficiency. PELT introduces a pruning rule that eliminates candidate change points that cannot be optimal in future iterations, therefore reducing the computational time while still providing an exact solution. These characteristics make this model a very good fit for the objectives of this thesis, and its formulation and implementation are further explained in 4.2.3.

Recent studies have registered considerable correlations and spillover effects in cryptocurrency markets. De Rosa et al. [2024] demonstrates very strong correlation patterns between Bitcoin and Ethereum against a list of 60 other cryptocurrencies, with the results showing very high correlation coefficients for some of the cryptos. These findings are particularly relevant for DOGE, as Huynh et al. [2020] shows that smaller market capitalization coins, including Dogecoin, are often receivers and senders of spillover effects with respect to other larger currencies.

One last important concept considered for this thesis was the identification of influential users in the subreddit. When analyzing networks with many nodes (which is the case of r/dogecoin with more than $400,000$ users/nodes), not all nodes have the same roles and relevance, and therefore identifying the most preponderant nodes is very significant (Lü et al. [2016]). Although there are several distinct ways of identifying subsets of influential nodes, this thesis follows an approach similar to Trimborn and Zwetsloot [2024], where the concept of "vital node" is defined and based not singularly on centrality measures but also on consistency and regularity of posting throughout the period of study.

The concepts and background materials explored in this section inspired the framework developed for the thesis, which is described in detailed in section 4.

3 Data processing

Each social media platform is organized in its own way, and Reddit is a perfect fit for this research due to its structure. Unlike, for example, Twitter, where users post on their profile and there is no separation of topics, Reddit is organized by individual communities named subreddits. Each user can join the subreddits they are interested in, and each subreddit has a specific theme. This makes the data collecting process much easier, since one can simply identify the relevant subreddit(s) for their research purposes and extract the data from there, ensuring the collected data is applicable.

3.1 Data Collection

For this thesis, data was collected from the r/dogecoin subreddit, which, as the name suggests, is dedicated specifically to discussions about Dogecoin. All interactions were fetched from the official Reddit API and the *Python Reddit API Wrapper (PRAW)* was used to facilitate the access, translating Python commands into the HTTP requests required by the Reddit API.

The data collection process was a challenge due to the Reddit API's policy changes and Terms of Service updates in the last years. After years of free and unlimited use, in 2023 Reddit started limiting user access to the API, implementing rate limits and removing several interactions from the database. PRAW implemented the necessary changes to comply with the new rules, which brought two essential issues: the limit of 100 queries per minute, and the fact that many posts and comments had already been deleted from the database. To solve the rate limit issue, a script was developed to cycle through all the posts,

extracting the comment tree in the most efficient way possible. In total, the data collection process took more than 1 month to complete using 2 computers in simultaneous. Regarding the deleted posts and comments, each time one showed up, the script automatically skipped it and moved on to the next interaction. This means not 100% of the original interactions was included in the data, however, the high amount of posts and comments collected still allowed for a very representative dataset.

3.2 Data structure

It is also important to mention the post structure. A user can submit a post on the subreddit, which can then be commented by other users. Users can comment either directly on the post itself, or on other people's comments, thus creating a tree of comments that can contain many layers. This allows for different ways of considering and interpreting the data. Taking the example from Figure 1, if user *Pr099* posts, user *Eden_Rich* comments on *Pr099*'s post, and then user *sexcitebike* comments on *Eden_Rich*'s comment, this can either be considered an interaction from *sexcitebike* to *Eden_Rich* (author of the comment that is being replied), or from *sexcitebike* to *Pr099* (author of the post).

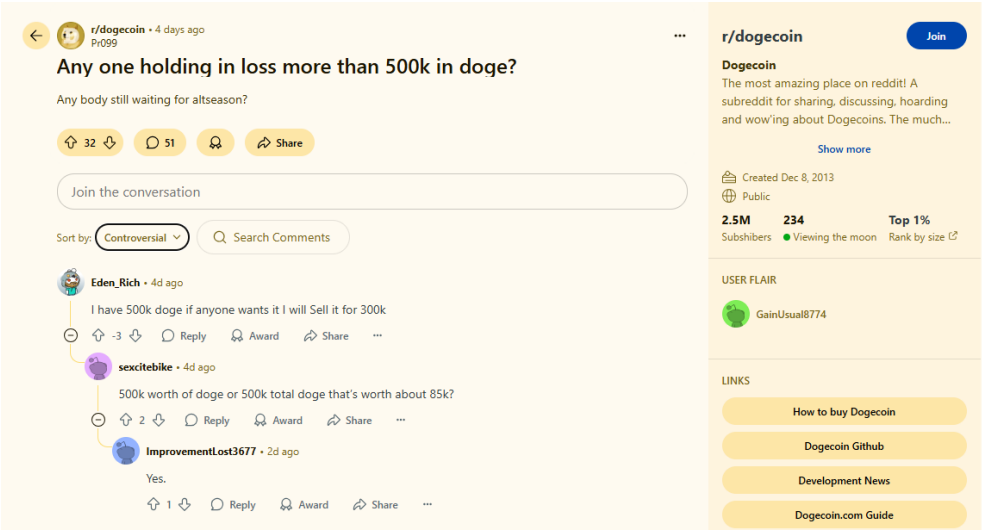

Figure 1: Structure of a Reddit post and its comment hierarchy.

In this particular research, all comments and subcomments are treated as comments on the original author of the post (e.g. in Figure 1, all 3 comments would count as a comment directed to user *Pr099*). This structure emphasizes the importance of nodes that write viral posts over nodes that comment on many viral posts and get several subcomments on their comments, thus resulting in a less sparse network focused on the influential posters. The relevance of this structure decision will become clearer in the next chapter of this article, when node importance is discussed.

The relevant variables that were fetched for each interaction are described in Table 1. Note that the title and content of the posts and comments are not collected, as the focus of this analysis is based on the quantity and timing of the interactions. Evaluating the content of these posts through techniques like sentiment analysis could be an interesting opportunity for further research.

Table 1: Description of the variables fetched from Reddit for each interaction

Variable Name	**Description**
`post.id`	Unique identifier for each Reddit post
`comment.id`	Unique identifier for each Reddit comment
`parent.id`	Unique identifier for the post or comment being replied to
`from`	Username of the author of the post or comment
`to`	Username being replied to (for comments; may match `from` for posts)
`time`	Timestamp of when the post or comment was created (local time)
`utc`	Timestamp in Unix epoch time (seconds since 1970-01-01 00:00:00 UTC)

Table 2 also explains some special cases of posts and comments that might not be fully clear when first interpreting the variables.

Table 2: Interpretation of Reddit Interaction Variables

Condition	**Interpretation**
`from` = `to` and `post.id` = `parent.id` = `comment.id`	User `from` published a new post on Reddit.
`from` = `to` and `post.id` = `parent.id` $\neq$ `comment.id`	User `from` commented on their own post.
`from` = `to` and `post.id` $\neq$ `parent.id` $\neq$ `comment.id`	User `from` published a sub-comment to their own previous comment in a post.
`from` $\neq$ `to` and `post.id` = `parent.id` = `comment.id`	User `from` commented directly on a post created by user `to`.
`from` $\neq$ `to` and `post.id` $\neq$ `parent.id` $\neq$ `comment.id`	User `from` published a sub-comment to a comment by user `to` in a post.

3.3 Data overview

Since the biggest Dogecoin frenzy occurred in early 2021, the analysis period was defined from the beginning of December 2020 to the end of July 2021, for a total of 8 months. This resulted in a total of 608,519 posts and 2,721,816 comments from 420,799 different users.

Figure 2 shows the distribution of posts and comments across the 8 months of analysis, next to the price of Dogecoin during the same period. There are some visual similarities between both plots, such as the first spike in end of January 2021 and the highest activity months of April and May 2021, which could be a positive first indication of the relevance of studying the social media activity related to Dogecoin.

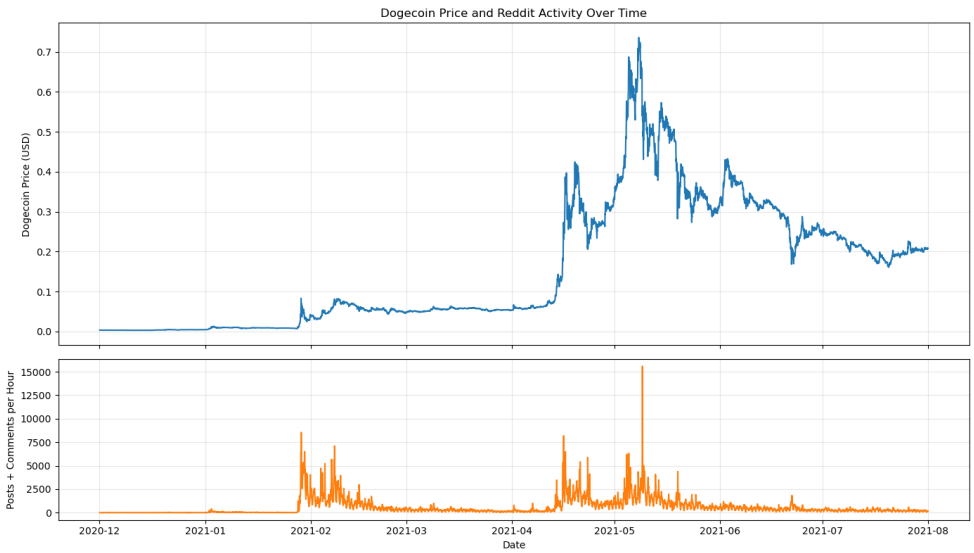

Figure 2: Dogecoin Price and Reddit Community Activity Over Time.

4 Methodology and model

After having structured the data, a modeling framework must be defined to build the bridge between a list of 3,000,000+ interactions and a measurable set of metrics that evaluates the Dogecoin market dynamics. This is achieved through the development of a framework which detects behavioral shifts in the community that can potentially anticipate market movements. After locating these moments of change in user behavior, there must also be a way of modeling them into actual market predictions. This thesis aims to provide a framework that addresses three essential points:

- Identification of a subset of influential users to reduce the complexity of the network, ensuring that the analysis focuses on meaningful and representative community members instead of attempting to model the behavior of all 400,000+ users.
- Application of a change point detection method, specifically, the Pruned Exact Linear Time (PELT) algorithm, to identify the moments in which the interactions with the influential users show significant behavior shifts in the community.

- Construction of predictive models for Dogecoin returns, trading volume, and volatility incorporating the identified change points, as well as relevant network metrics and the lags for all variables, therefore quantifying the relationship between social activity and market dynamics.

4.1 Identifying influential nodes

The first step after the data collection process is to define the set of users to be included in the model estimations. Working with the full set of 400,000+ nodes does not make sense, since most of these users are not very active and lack enough interactions over the 8-month period of analysis to allow for meaningful results. As can be observed in Table 3, only 13.07% of users submit at least 20 comments or posts. The high number of total nodes also means that studying the relationships between pairs of nodes might be hard: only 3,899 nodes (0.93% of the total) have more than 10 direct interactions with at least one other node. This sparsity in the network increases the difficulty of analyzing relationships between two specific nodes due to the lack of data for most pairs of nodes.

Table 3: Summary of Reddit User Participation Metrics

Category	**Count**	**Percentage of Total**
Users with interactions	420,799	100.00%
Users with at least 5 interactions	164,317	39.05%
Users with at least 20 interactions	55,005	13.07%
Nodes in a pair with >10 interactions	3,899	0.93%

Therefore, a subset of nodes for the analysis must be defined, including only the nodes that are influential in the network and have a relevant number of interactions. These influential nodes are commonly referred to in literature as Vital Nodes (VN).

4.1.1 The (VN - ANVN) approach

There can be different approaches to the definition of vital nodes. Lü et al. [2016] performed a comprehensive analysis on the relevance of vital nodes and presented different vital node identification algorithms based on network analysis metrics. However, the concept of vital node can vary a lot depending on the application, and therefore this thesis follows a data driven approach that focuses on users that not only have many interactions but also post consistently throughout the 8-month period.

Therefore, a first approach considered to deal with node importance was to replicate the definitions used in Trimborn and Zwetsloot [2024]. The first definition establishes the concept of vital node.

Definition 1. *A user in a network is a* ***vital node*** *if it submits posts in at least 10% of the weeks and receives the highest 10% median comments on their posts.*

This definitions establishes vital nodes as users that are not only active and regular posters (through a minimum number of different weeks posting) but also get a high amount of comments on their posts (through the median comments percentile). Since the research by Trimborn and Zwetsloot [2024] is based on a model that uses node pairs to study interactions between two specific nodes, it also defines a second category of nodes to include in the pairs with vital nodes, since including all 498,979 nodes would result in too many pairs of nodes, and therefore a sparse and very computationally expensive model. These are called the admissible non-vital nodes (ANVNs).

Definition 2. *A user in a network is an* ***admissible non-vital node (ANVN)*** *of vital node i when that node has posted at least 4 comments in the posts of vital node i.*

Considering only the (VN - ANVN) pairs of nodes for the model, instead of (VN - every node) pairs, considerably increases the efficiency of the model by removing the nodes that

do not have enough interactions with vital nodes, for which changes in behavior would be hard to identify.

These definitions of vital nodes and admissible non-vital nodes were then tested in the r/dogecoin data to evaluate the feasibility of replicating the same change point detection model used for the r/wallstreetbets data, however, the distribution of the interactions between the users was considerably different, as seen in Table 4.

Table 4: Reddit Community Structure: r/wallstreetbets (filtered for GME) vs. r/dogecoin

	r/wallstreetbets (GME)	**r/dogecoin**
Total users who post/comment	498,979	420,799
Total VNs	64	298
Total ANVNs	48,213	6,079
Total VN-ANVN pairs	51,245	6,321

While the total number of active users is rather similar, the definitions used identify over 4 times more vital nodes in the Dogecoin community, indicating that the users with the most viral posts (in the sense of getting a lot of comments) might have more consistency and post more often across time, while the most influential users' contribution on the r/wallstreetbets community was more ephemeral.

However, the biggest difference stands out by looking at the number of admissible non-vital nodes. Despite having 4 times more vital nodes, which could imply a higher number of ANVNs, the Dogecoin community only gets 6,321 ANVNs, much less than the 48,213 ANVNs in r/wallstreetbets. When counting the total number of (VN-ANVN) connections, this results in more than 8 times less pairs to evaluate the model on.

This major contrast between the datasets, in which the Dogecoin dataset did not provide enough meaningful relationships between nodes, motivated the adoption of a different methodology that better suited the structure of the community and behavior of the users.

4.1.2 The Vital Node (VN) redefinition

Looking at the comparison shown in Table 4, one can infer that the higher number of VNs is a result of a more scattered distribution of posts throughout the period of analysis. Table 5 shows that the most influential nodes (measured by total comments received) in the community are actually active over a large period of time, with all of them posting in at least 10 different weeks, and 6 of them posting in more than half of the weeks in the 8-month period.

Table 5: Top 10 Vital Nodes (VNs) with most total comments received

Username	**Weeks Active**	**Total Comments**
42points	27	116,176
iUsedToCallDogeDodge	19	9,231
rizzo49er	16	6,860
dongnyoung1	10	6,785
Temporary-Muffin-756	25	6,113
raddit13	26	5,145
mrsmile420	25	4,841
popcity_peep	23	4,586
DogeIsMyPower	14	4,188
ericqwaqwa	11	3,949

Therefore, a new approach was defined. Instead of modeling each relationship between a set of pairs of nodes, the new structure proposes dropping the concept of admissible non-vital node, focusing only on the vital nodes and ignoring the identity of the user that posted the comments, counting all interactions from different users as one value. This structure has some considerations:

- Instead of creating a time series for every (VN - ANVN) pair, only one time series is created per VN. This means each time series has considerably more data points: it not only removes the need for filtering for ANVN comments, therefore integrating all comments from all users, but also it aggregates all the comments into one count/time series.

- While the time series have more data, there are considerably less time series to evaluate the model on (only one per VN). To account for this reduction, the vital node definition was relaxed, resulting in a higher number of vital nodes.

- The model approach presented by Trimborn and Zwetsloot [2024] is a complex formulation built not only on the number of interactions between nodes but also including centrality and network activity measures. With the restructuring applied on this thesis, the followed approach became a much simpler method, based on a traditional change point detection algorithm. While this could potentially have had a negative influence on the quality of predictions, the results in the end showed that the performance of the simpler model were still considerably accurate, while being much faster and computationally cheaper.

Based on this factors, the new structure consisted of the elimination of the admissible non-vital node concept, and the redefinition of vital node.

Definition 3. *A user in a network is a* ***vital node*** *if it submits posts in at least 8 different weeks and receives the highest 25% median comments on their posts.*

This new definition was based on two key changes: increasing the number of minimum weeks posting to get more scattered time series, and lowering the percentile of highest median comments to increase the total number of vital nodes, in order to get more time series to evaluate. The robustness of the definition was also evaluated and included in Section 7.

Redefining the vital node requirements resulted in a total of 1,066 vital nodes with a median of 248 comments per vital node, allowing for a meaningful analysis in each of the time series. As seen in Figure 3, the distribution of posts by vital nodes and comments under those posts along the 8-month period is visually similar to the distribution of all posts and

comments, suggesting that the subset of the most influential nodes can be representative of the whole network.

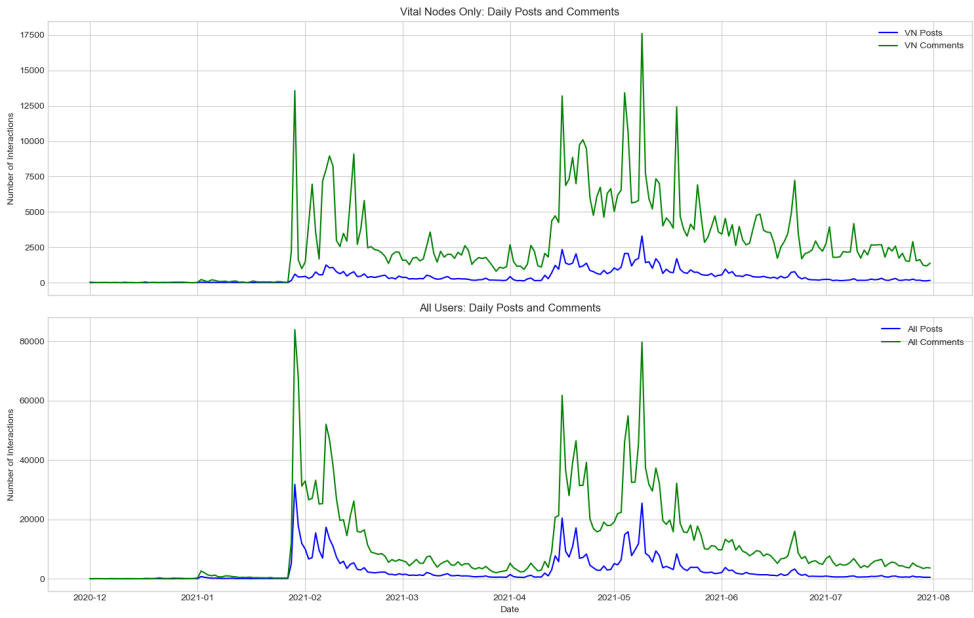

Figure 3: Interaction timeline: All users vs. VN only.

In general, the structure of the dataset motivated a change in the methodology, which resulted in a simpler and more efficient change point detection framework, while still producing relevant results, as will be shown in Section 5.

4.2 Change point detection method

After identifying the relevant subset of vital nodes, an hourly time series is built for each VN with the number of comments received on their posts. The goal is to develop a framework to detect changes in user behavior through the analysis of the time series. The approach chosen to tackle this challenge consists of a change point detection methodology.

Change point detection is a statistical technique that identifies time points in which the properties of a time series, such as the mean, variance, or trend, change significantly. The idea is essentially to partition the time data into segments, where each segment shows

consistent behavior within its data points and adjacent segments differ. The points that separate these segments are the change points, and the algorithms usually go through a set of admissible changepoints and select the ones that best isolate the segments by minimizing a defined loss function. This method has been widely adopted and applied to several fields such as health, finance, and social network analysis.

4.2.1 Computational considerations

There are various strategies and algorithms to detect change points in a time series. Bai et al. [2021] proposes an approach that locates change points in high-dimensional Vector Autoregressive (VAR) models with sparse transition matrices. This approach is adopted in Trimborn and Zwetsloot [2024] for detecting change points in the r/wallstreetbets users' behavior, thus showing its effectiveness in the social network modeling context. This method is particularly strong when dealing with high-dimensional data and it successfully handles multivariate time series with dependencies across many variables. These factors make it a suitable framework for studying the complex Reddit community networks with thousands of users. It does, however, come with a cost. Being a much more complex model, it is also very computationally expensive, particularly with very large datasets. Even with the filtering of vital nodes and reduction of total relationships to model, it is still an extremely complex computation.

With this in mind, this thesis proposes a simpler solution that significantly improves the computation time. While the model presented in Bai et al. [2021] allows for a more comprehensive approach, such as the inclusion of other network features and lagged variables, the selection of a simpler model opens the door for a much faster analysis, based on univariate (the number of comments received by a VN in an hour) time series, which can even be seen as an exploratory framework to evaluate viability and robustness before diving into more complex modeling.

4.2.2 Detecting change points with a low computational cost

In general, the challenge of detecting multiple change points is formulated as a minimization of a penalized cost function summation over the segmentations created by the possible sets of changepoints:

$$\min_{m,\,\tau_1,\ldots,\tau_m} \left\{ \sum_{i=0}^{m} C\left(y_{(\tau_i+1):\tau_{i+1}}\right) + \beta m \right\} \tag{3}$$

where:

- m is the number of changepoints,
- τ_i are the changepoint locations ($\tau_0 = 0$, $\tau_{m+1} = n$);
- $C(\cdot)$ is a segment cost function;
- β is a penalty parameter;

There are several methods to detect multiple change points. Binary Segmentation (BS) is a recursive method that generalizes a method to find a single change point through the equation 2, minimizing 4.2.2 (specifically with $f(m) = m$). It starts by looking for a single change point, and then iteratively applies the equation to each new segment generated by the previous changepoints, until no significant new change points are found in any segment. The advantage of BS is its computational efficiency, resulting in a $\mathcal{O}(n \log n)$ calculation. However, this is just an approximate method, and therefore it is not guaranteed to find the global minimum of 4.2.2.

While BS brings computational efficiency as an advantage, an exact method that always finds the optimal partition of segments according to the cost function is preferred. Jackson

et al. [2004] proposed the Optional Partitioning (OP) method, based on dynamic programming. It computes the minimization 2 for $t = 1, ..., n$ (starting on $t = 1$) sequentially, thus providing a recursion which gives the optimal cost for data $y_{1:s}$ in terms of the optimal cost for data $y_{1:t}$ for $t < s$. The exactness of this method comes, however, with a setback for large datasets. The cost of solving a recursion for time s is linear in s, therefore the total computational cost to find $F(n)$ is quadratic in n.

4.2.3 Pruned Exact Linear Time (PELT) Method

To combine computational efficiency with an exact method, Killick et al. [2012] proposed the Pruned Exact Linear Time (PELT) method, which is based on the Optimal Partitioning method but adds a pruning rule that increases the computational efficiency of OP. The concept of pruning is to remove the values of τ that can never be optimal according to the minimization performed at each iteration of the OP algorithm. The following theorem gives a condition that allows for the desired pruning:

Theorem 1. *We assume that when introducing a changepoint into a sequence of observations the cost, C, of the sequence reduces. More formally, we assume there exists a constant K such that for all $t < s < T$,*

$$C\left(y_{(t+1):s}\right) + C\left(y_{(s+1):T}\right) + K \leq C\left(y_{(t+1):T}\right). \tag{4}$$

Then if

$$F(t) + C\left(y_{(t+1):s}\right) + K \geq F(s) \tag{5}$$

holds, at a future time $T > s$, t can never be the optimal last changepoint prior to T.

The proof of this theorem is in 8.1. The pruning rule can easily be added to the OP algorithm, and it considerably increases the computational efficiency of the algorithm.

The PELT algorithm was implemented in Python through the `ruptures` library, and it was run with the following parameters:

- 168h (1 week) minimum time between change points;
- $\beta = 2$ penalization parameter;
- $C(y_{a:b}) = \sum_{t=a}^{b}(y_t - \bar{y}_{a:b})^2$ cost function, with $\bar{y}_{a:b} = \frac{1}{b-a+1}\sum_{t=a}^{b} y_t$.

The minimum time of 1 week between the change points facilitates the model estimation. Since a model with good forecasting performance is desired, the Akaike Information Criterion, AIC (Akaike [1974]) penalization parameter β was defined as $\beta = 2$ (the AIC criterion penalty parameter corresponds to $\beta = 2p$, where p is the number of parameters per segment of the cost function, and in this context $p = 1$). The robustness of the penalty parameter selection was evaluated and included in Section 7. The cost function used is the $L2$ cost (sum of squared errors), which is defined as the sum of squared deviations from the segment mean. Therefore, it generates the optimal set of change points by detecting shifts in the mean of the segments. One advantage of the $L2$ function is its considerable computational efficiency, which perfectly fits the goal of this thesis.

The results of the change point estimation are presented in the next section 5.

4.3 Leveraging change points to model market dynamics

After detecting the change points for each vital node, a model must be defined to predict the market dynamics of Dogecoin. This is done with hourly frequency, including both the change points previously estimated and hourly Dogecoin data extracted from Binance. Three essential metrics were estimated:

- Return (specifically log-return), calculated from the dataset through the closing price, $Return_t = log(\frac{Close_t}{Close_{t-1}})$;

- Trading Volume (in USD), directly extracted from Binance;
- Volatility, since only hourly data was available, it was calculated as Realized Volatility through the squared log-return, $Volatility_t = (Return_t)^2$.

The model definitions were then established for the 3 metrics:

$$Return_t = \alpha + \sum_{k=1}^{K} \beta_k \, Return_{t-k} + \sum_{k=0}^{K} \gamma_k \, CPs_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{6}$$

$$TV_t = \alpha + \sum_{k=1}^{K} \beta_k \, TV_{t-k} + \sum_{k=0}^{K} \gamma_k \, CPs_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{7}$$

$$Volatility_t = \alpha + \sum_{k=1}^{K} \beta_k \, Volatility_{t-k} + \sum_{k=0}^{K} \gamma_k \, CPs_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{8}$$

The models predict returns, trading volume and volatility using:

- Their own past values (lagged variables);
- The number of change points detected (lagged as well);
- Network Activity, which corresponds to the total number of interactions on r/dogecoin during hour $t - k$.

The metrics are estimated through a rolling window analysis. Models 4.3, 4.3, and 4.3 are applied in a window of 1 month (so $K = 719$, since 1 month corresponds to $30 \times 24 = 720$ hours), therefore not including a forecast for the first month of data (December 2020). The model is then fitted and the parameters are estimated through a linear regression using the data from the window. The window then moves forward by a predefined step size of 24 hours, and the regression process is repeated for the new window. This process continues until the window reaches the end of the data.

The following section 5 contains the results of the rolling window estimation.

5 Results/Analysis

After thoroughly explaining the framework for this thesis in the previous section, the methods are applied to the Dogecoin dataset between 01/12/2020 and 31/07/2021 to evaluate if the behavior of r/dogecoin users has an influence on Dogecoin market behavior. First, the change point detection method is run for every vital node, outputting a set of change points corresponding to that VN. The change points of all the vital nodes are then combined into a data frame that has the total number of change points per hour in the network for the 8-month period. Finally, these changepoints, together with the DOGE market data extracted from Binance and the Network Activity measure, are used to estimate the Return, Trading Volume and Volatility of Dogecoin. This process is shown in more detail in the rest of this section.

5.1 Change point detection

As previously mentioned in section 4.2, the change point detection algorithm was run with the following parameters:

- Vital nodes corresponding to the 75th percentile of highest median comments received per week and posting in at least 8 different weeks;
- L2 cost function

Figure 4 shows most vital nodes end up having between 2 to 6 change points. There are, however, some outliers with many changepoints. User `42points`, the creator of r/dogecoin and arguably the most influential user (as observed in Table 5), has a total of 21 change points. This shows a tendency for users with more interactions to display more change points, due to having more data in the time series of interactions.

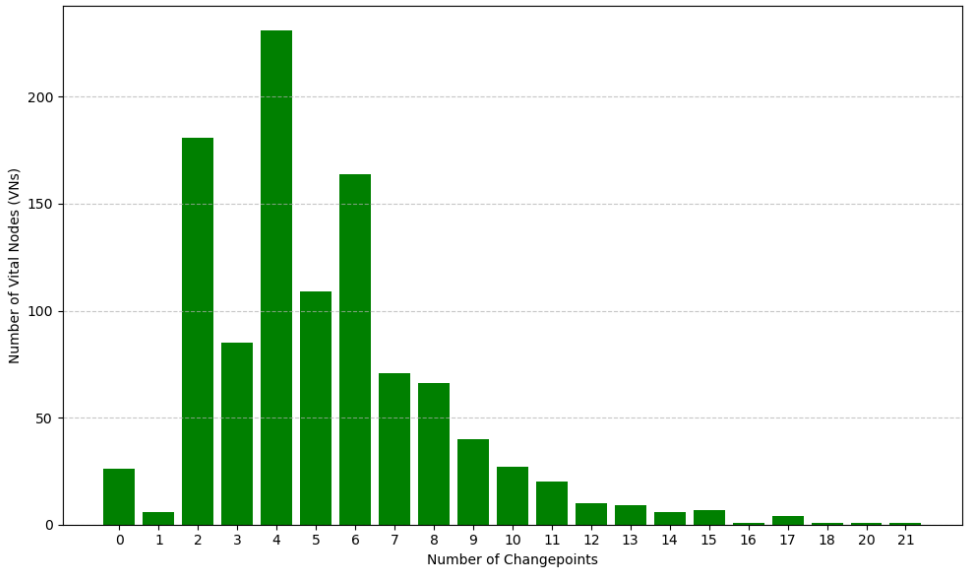

Figure 4: Distribution of change points per vital node.

Looking at the distribution of the change points in Figure 5 and its comparison with DOGE market metrics in Figure 6, there are clear visual similarities in the spikes of all 3 plots, with some change point steep increases even coming before the increases in the market metrics, as seen in the 2 identified periods in Figure 6 in end of January 2021 and early April 2021.

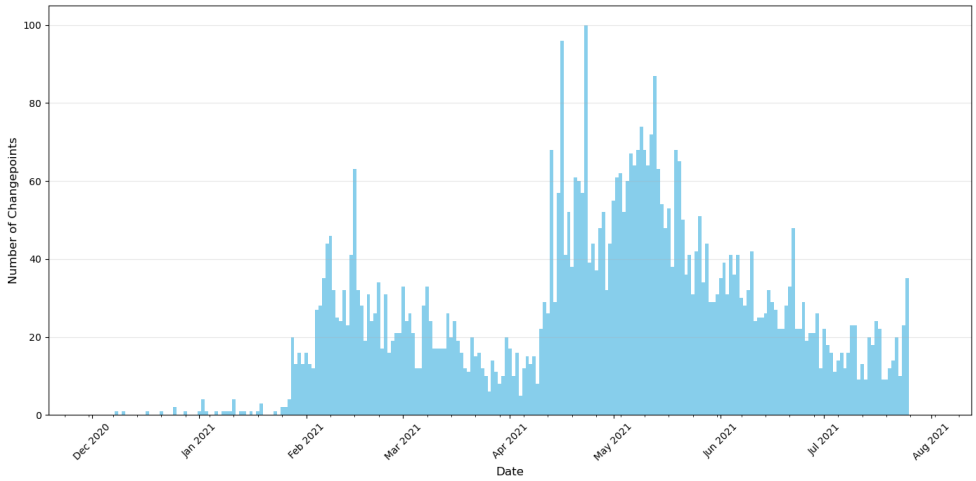

Figure 5: Distribution of change points between December 2020 and July 2021.

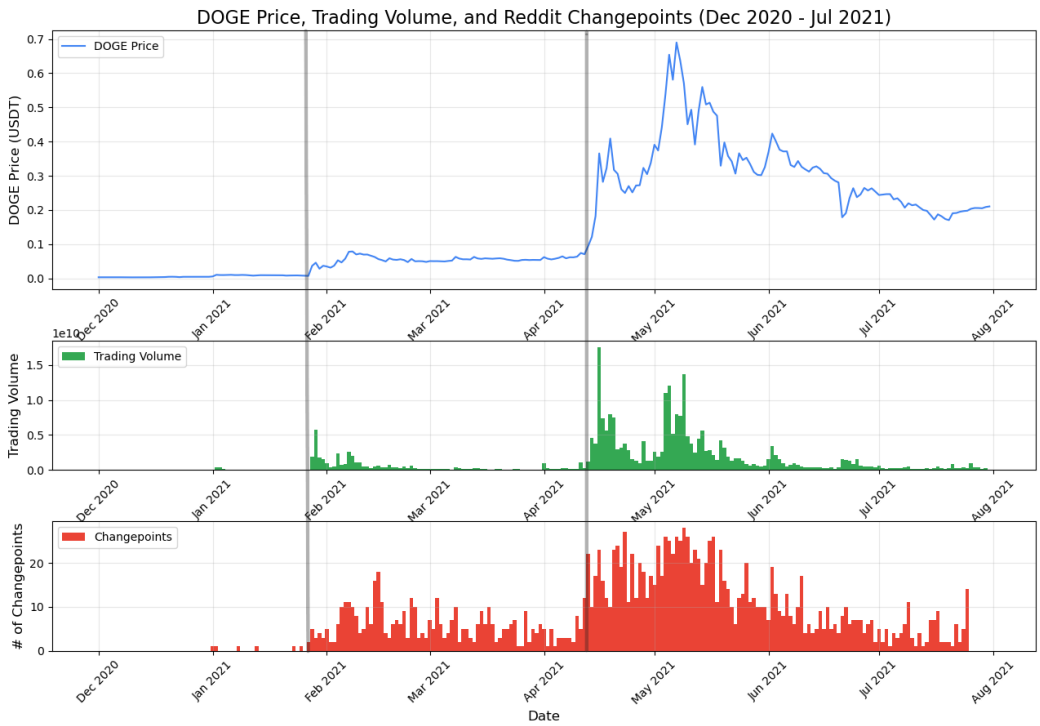

Figure 6: DOGE price and trading volume compared to change points.

5.2 Market metrics model estimation

After getting the set of change points from the PELT model, the next step was to model the market metrics. To start, models without the change points, consisting of just the market activity and the lagged variables was defined as the baselines:

$$Return_t = \alpha + \sum_{k=1}^{K} \beta_k \, Return_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{9}$$

$$TV_t = \alpha + \sum_{k=1}^{K} \beta_k \, TV_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{10}$$

$$Volatility_t = \alpha + \sum_{k=1}^{K} \beta_k \, Volatility_{t-k} + \sum_{k=0}^{K} \eta_k \, NetworkActivity_{t-k} + \epsilon_t \tag{11}$$

These models were then compared to the models 4.3, 4.3, and 4.3 to evaluate the impact of change points in Dogecoin market dynamics and whether it adds anything to the network activity data.

The findings show that, overall, the network activity model equations, even without the change points, are already considerably correlated to the evaluated market metrics. Trading volume was particularly well estimated by the models, with the adjusted R^2 values going above 0.9 in some days.

However, it is also important to mention that adding the change points also improves the model's explanatory power. This improvement is particularly relevant in the return predictions. Besides that, across all metrics, the addition of change points is especially

pronounced in periods of high market activity, therefore suggesting that the change point identification might anticipate the spikes and contribute more to the model performance in those periods.

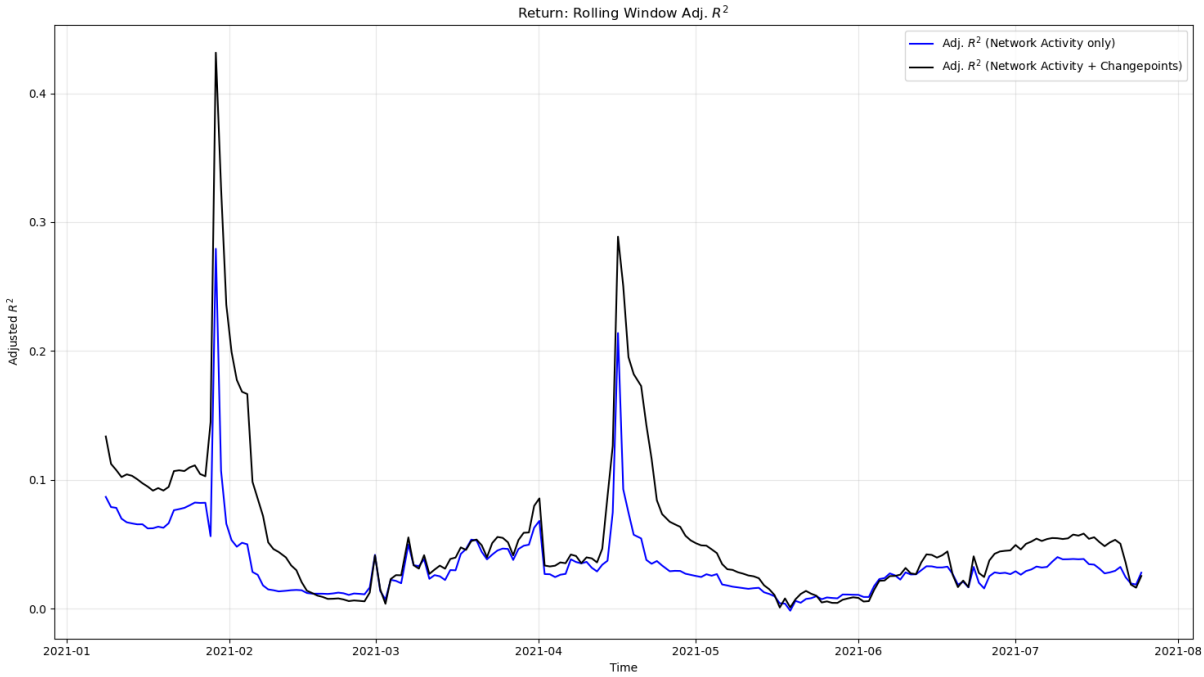

Figure 7: Return modeling.

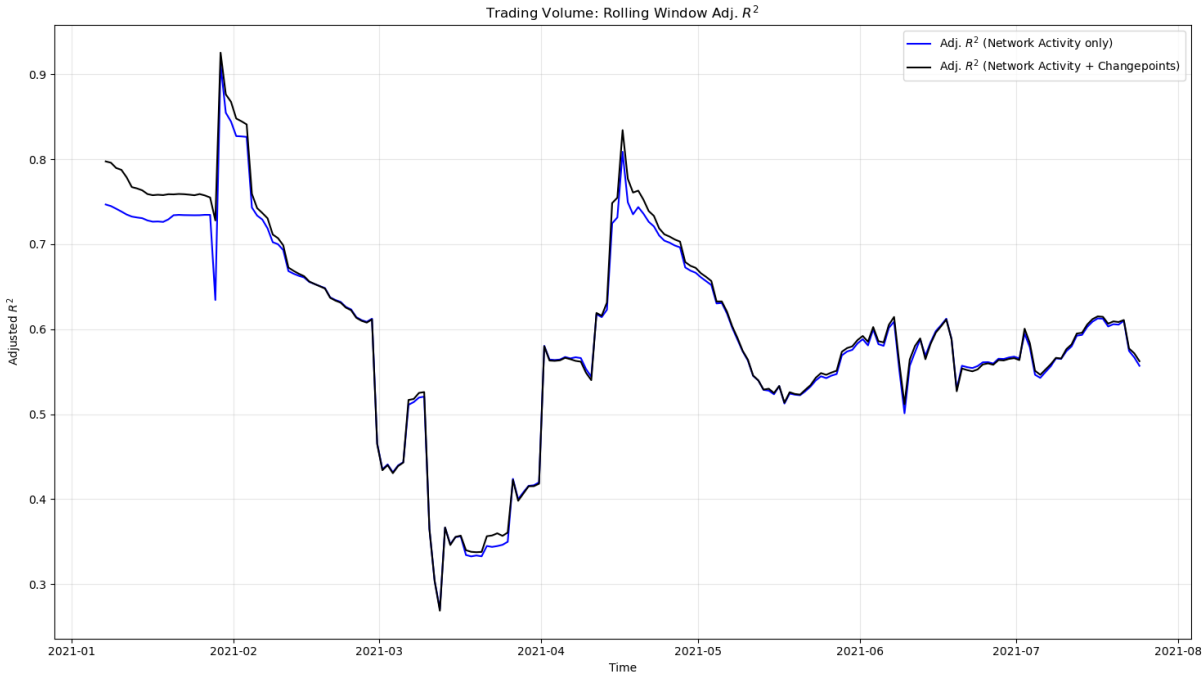

Figure 8: Trading volume modeling.

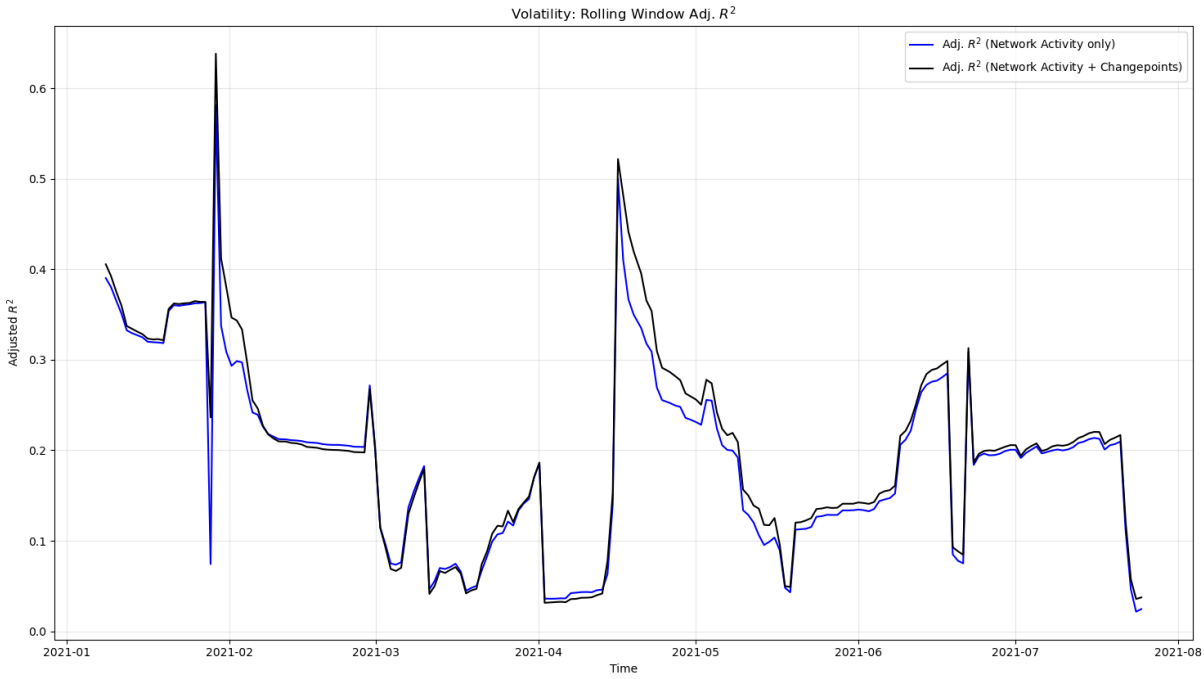

Figure 9: Volatility modeling.

Looking at the modeled market metrics on Figures 7, 8, and 9, several insights can be drawn.

The return model 7 has a relatively low baseline predictability, with adjusted R^2 values mostly below 0.1, indicating that returns are inherently hard to predict, which is something commonly observed in the financial modeling field and corresponds to the expectations. However, the addition of change points to the model has a very positive impact, significantly improving the model performance and showing that change point analysis can add even more context than just looking at network activity. Also, around the outbursts in DOGE in end of January and early April, a spike can also be seen in the adjusted R^2, suggesting a stronger performance in the periods of higher activity.

The trading volume model 8 has different characteristics. Unlike the return model, it displays a very high baseline model performance, with adjusted R^2 values consistently above 0.5, and even reaching above 0.9 and 0.8 around the January and April outbursts, respectively. This suggests that the r/dogecoin network activity (corresponding to the total number of interactions per hour) is already a very strong predictor of the trading volume. Once again, around the periods of more activity, the model performs particularly well, and it is also in these periods that the change points improve on the baseline performance, while during the rest of the 8-month period they do not add much to the baseline model.

Finally, looking at the volatility model 9, the results are more balanced when compared to return and trading volume. The baseline model performs better than the return baseline model, but worse than the trading volume baseline model. The change points do not have an extremely significant impact on the model predictability, however, the model with the change points performs consistently above the baseline, which even if the difference is not very large, still indicates that the change points are a meaningful addition to the network activity predictions.

In summary, the model with change points performs better than the baseline network

activity model, thus proving its effectiveness and the relevance of identifying shifts in behavior through change point detection methods to anticipate market movements. All the models also perform particularly well around the periods of highest activity, especially the end of January and early April in 2021. This shows the power of social networks models, which are even more successful in periods of high speculation and uncertainty.

5.3 Analyzing r/Dogecoin community analysis spillover on Bitcoin

While the main analysis of this thesis focuses on the relationship between the r/Dogecoin community and the DOGE market dynamics, it is also relevant to analyze the spillover effect of the analysis on other cryptocurrencies, and for that purpose Bitcoin - the biggest cryptocurrency in the world - was selected.

The approach was to use the model definitions defined in section 4 to estimate the return, trading volume, and volatility of Bitcoin using the lagged BTC metrics, while using the network activity and change points detected for r/Dogecoin. The results are presented in tables 5.3 5.3 5.3.

Looking at the obtained R^2 and Adjusted R^2, and in line with the results from section 5, the Trading Volume exhibits considerably higher predictability than Return and Volatility, confirming that it is the most predictable metric when using social community interactions data. Figure 10 shows that, even though the patterns in trading volume between the two assets are not equal, there are certain time periods, particularly after April 2021, where there are spikes in Trading Volume for both DOGE and BTC around similar days, which could explain the relatively high R^2. The results were, however, not as high for Bitcoin as for Dogecoin, which makes all sense since the community in question is dedicated to Dogecoin.

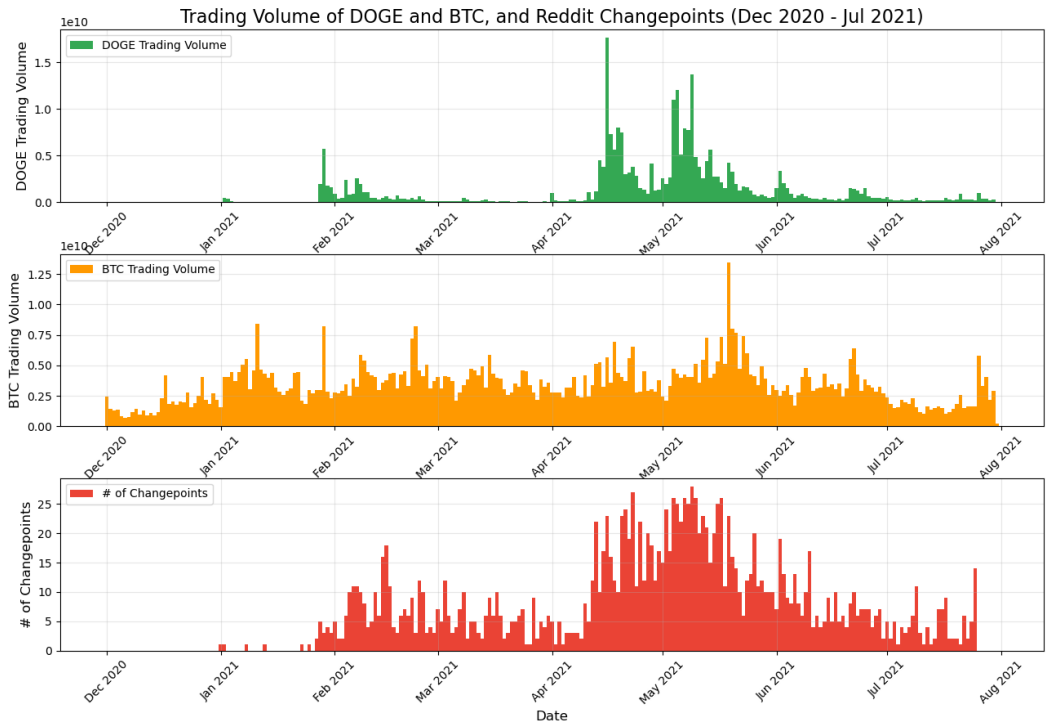

Figure 10: DOGE and BTC trading volume compared to change points.

The Return table 5.3 yields very interesting insights. While the model performance is overall quite low for both DOGE and BTC return (even slightly better for BTC), adding the change points to the models improves the DOGE model performance way more than it improves the BTC model performance. This suggests that while BTC might show some sensitivity to the network activity from r/Dogecoin, the implementation of the behavior shift detection through change point identification is way more impactful within the context of DOGE.

In summary, the spillover from r/Dogecoin community indicators to Bitcoin is, overall, not considerably noticeable. Nevertheless, the results show that Bitcoin trading volume can still be somewhat predicted through Dogecoin Reddit activity, achieving decent R^2 values, which illustrates the correlation and connectedness of social media driven decisions in markets. For returns, the spillover exists mostly through the baseline network activity

rather than through behavioral shift detection - the change points add much more context for Dogecoin itself. This supports the conclusion that collective behavioral shifts in the r/Dogecoin community drive market reactions with limited effects on other assets (in this case BTC), reinforcing the importance of isolating social media analysis for each crypto asset.

Table 6: Return model performance for Dogecoin and Bitcoin

Cryptocurrency	Network Activity	Network Activity + VNs
Dogecoin	0.0050 / 0.0025	0.0102 / 0.0061
Bitcoin	0.0054 / 0.0028	0.0070 / 0.0030

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

Table 7: Trading volume model performance for Dogecoin and Bitcoin

Cryptocurrency	Network Activity	Network Activity + VNs
Dogecoin	0.6928 / 0.6920	0.6963 / 0.6950
Bitcoin	0.4706 / 0.4692	0.4733 / 0.4711

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

Table 8: Volatility model performance for Dogecoin and Bitcoin

Cryptocurrency	Network Activity	Network Activity + VNs
Dogecoin	0.1893 / 0.1872	0.1912 / 0.1879
Bitcoin	0.0509 / 0.0484	0.0526 / 0.0488

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

6 Conclusion

This thesis set the initial objective of understanding if Reddit user behavior had an impact on Dogecoin market dynamics, specifically by detecting abrupt shifts in activity

around influential members of the community, and using them to model price metrics.

The empirical findings show that the change point detection does in fact add value to the predictive value of the baseline network activity models, especially when modeling return. The improvements are particularly notable during periods of high speculation and market activity, which further highlights the relevance of behavioral shifts for predicting periods of agitation. In parallel, the behavioral shifts in the Dogecoin Reddit community display only limited spillover to Bitcoin, with network activity proving to be a better cross-predictor than the change points, which are more tied to Dogecoin specifically.

The model results show that, while network activity can already explain the trading volume quite well, with adjusted R^2 reaching above 0.9 in certain periods, the inclusion of change points enhances the models, especially for return and volatility. It is also important to underline the efficiency of the model, which allows for a quick, yet accurate analysis.

There are, obviously, points for improvement and opportunities for further research. The analysis was limited to Dogecoin and Reddit, but there are more meme assets that can be studied, such as meme stocks or other cryptocurrencies, as well as other social networks that could be used to model user interactions. There is also space for more comprehensive modeling, with the possibility of including more variables in the models, such as sentiment analysis or Dogecoin news, which could enrich the model and improve its predicting capability. Finally, the PELT model built for this thesis presents an extremely efficient model, which opens the door for more complex solutions that can adapt to the data even better.

In summary, this thesis shows the potential of behavioral analysis in highly volatile and socially driven markets, particularly highlighting the value of group shifts of behavior as a powerful analytical tool.

7 Robustness evaluation

7.1 Vital node definition robustness

The robustness checks across different possible definitions of vital nodes are in line with the results obtained in Section 5. Return models have the lowest values of R^2 and adjusted R^2 among the 3, however, the addition of the change points of the vital nodes considerably increases the performance of the model. On the other hand, Trading Volume models have a great baseline performance, but the change point analysis does not add much to the network activity model. Overall, the different definitions provide relatively similar R^2 and adjusted R^2, thus showing the robustness of the chosen VN definition.

Table 9: Return model performance across Vital Node (VN) Definitions

VN Definition	Network Activity	Network Activity + VNs
75% median, 5w	0.0050 / 0.0025	0.0102 / 0.0061
75% median, 8w	0.0050 / 0.0025	0.0102 / 0.0061
75% median, 10w	0.0050 / 0.0025	0.0104 / 0.0064
85% median, 5w	0.0050 / 0.0025	0.0101 / 0.0060
85% median, 8w	0.0050 / 0.0025	0.0092 / 0.0052
85% median, 10w	0.0050 / 0.0025	0.0087 / 0.0046

Note: Each cell reports R^2 / Adjusted R^2 for the full sample. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns additionally use changepoint counts (and their lags) for vital nodes.

Table 10: Trading Volume model performance across Vital Node (VN) Definitions

VN Definition	Network Activity	Network Activity + VNs
75% median, 5w	0.6928 / 0.6920	0.6982 / 0.6970
75% median, 8w	0.6928 / 0.6920	0.6963 / 0.6950
75% median, 10w	0.6928 / 0.6920	0.6951 / 0.6939
85% median, 5w	0.6928 / 0.6920	0.6976 / 0.6964
85% median, 8w	0.6928 / 0.6920	0.6957 / 0.6945
85% median, 10w	0.6928 / 0.6920	0.6950 / 0.6938

Note: Each cell reports R^2 / Adjusted R^2 for the full sample. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns additionally use changepoint counts (and their lags) for vital nodes.

Table 11: Volatility model performance across Vital Node (VN) Definitions

VN Definition	Network Activity	Network Activity + VNs
75% median, 5w	0.1893 / 0.1872	0.1921 / 0.1888
75% median, 8w	0.1893 / 0.1872	0.1912 / 0.1879
75% median, 10w	0.1893 / 0.1872	0.1911 / 0.1878
85% median, 5w	0.1893 / 0.1872	0.1920 / 0.1887
85% median, 8w	0.1893 / 0.1872	0.1923 / 0.1890
85% median, 10w	0.1893 / 0.1872	0.1915 / 0.1882

Note: Each cell reports R^2 / Adjusted R^2 for the full sample. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns additionally use changepoint counts (and their lags) for vital nodes.

7.2 Penalty parameter robustness

Testing alternative penalty parameters values for the change point detection further validates the stability and robustness of the approach and methodology. Although alternatives from the AIC criterion can be considered - in this case the BIC criterion and a higher value of $\beta = 20$ - overall the AIC penalty has an edge over the other β values. Still, the R^2 and adjusted R^2 values stay relatively consistent between all 3 values, once again showing the robustness of the method.

Table 12: Return model performance across different penalty parameters β

β	Network Activity	Network Activity + VNs
$\beta = 2$ (AIC)	0.0050 / 0.0025	0.0102 / 0.0061
$\beta \approx 8.66$ (BIC)	0.0050 / 0.0025	0.0088 / 0.0047
$\beta = 20$	0.0050 / 0.0025	0.0082 / 0.0037

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

Table 13: Trading Volume model performance across different penalty parameters β

β	Network Activity	Network Activity + VNs
$\beta = 2$ (AIC)	0.6928 / 0.6920	0.6963 / 0.6950
$\beta \approx 8.66$ (BIC)	0.6928 / 0.6920	0.6966 / 0.6953
$\beta = 20$	0.6928 / 0.6920	0.6903 / 0.6889

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

Table 14: Volatility model performance across different penalty parameters β

β	Network Activity	Network Activity + VNs
$\beta = 2$ (AIC)	0.1893 / 0.1872	0.1912 / 0.1879
$\beta \approx 8.66$ (BIC)	0.1893 / 0.1872	0.1910 / 0.1877
$\beta = 20$	0.1893 / 0.1872	0.1914 / 0.1877

Note: Each cell shows R^2 / Adjusted R^2. "Network Activity" columns use lagged network activity and lagged returns as predictors. "Network Activity + VNs" columns also use changepoint counts (and their lags) for vital nodes.

8 Appendix

8.1 Pruning rule (proof)

Proof. Assume that

$$F(t) + C\left(y_{(t+1):s}\right) + K \geq F(s) \tag{12}$$

is true. Then

$$F(t) + C(y_{(t+1):s}) + \beta + K \geq F(s) + \beta$$

$$\implies F(t) + C(y_{(t+1):s}) + \beta + K + C(y_{(s+1):T}) \geq F(s) + \beta + C(y_{(s+1):T})$$

$$\implies F(t) + C(y_{(t+1):T}) + \beta \geq F(s) + \beta + C(y_{(s+1):T}),$$

according to

$$C\left(y_{(t+1):s}\right) + C\left(y_{(s+1):T}\right) + K \leq C\left(y_{(t+1):T}\right). \tag{13}$$

Hence, it follows that t cannot be a future minimizer of the sets

$$S_T := \left\{F(\tau) + C(y_{(\tau+1):T}) + \beta,\ \tau = 0, 1, \ldots, T-1\right\}, \quad T > s$$

and can be removed from the set of τ for each future step. □

8.2 Linear computational cost of PELT

Theorem 2. *Let θ^* be the value that maximizes the expected log-likelihood:*

$$\theta^* = \arg\max_{\theta} \int \int f(y|\theta) f(y|\theta_0) dy\, \pi(\theta_0) d\theta_0.$$

Let θ_i be the true parameter associated with the segment containing y_i and $\hat{\theta}_n$ be the maximum likelihood estimate for θ given data $y_{1:n}$ and an assumption of a single segment:

$$\hat{\theta}_n = \arg\max_{\theta} \sum_{i=1}^{n} \log f(y_i|\theta).$$

Then if

(A1) Denoting

$$B_n = \sum_{i=1}^{n} \left[\log f(y_i|\hat{\theta}_n) - \log f(y_i|\theta^*)\right],$$

we have $\mathbb{E}(B_n) = o(n)$ and $\mathbb{E}\left([B_n - \mathbb{E}(B_n)]^4\right) = \mathcal{O}(n^2)$;

(A2) $\mathbb{E}\left([\log f(Y_i|\theta_i) - \log f(Y_i|\theta^*)]^4\right) < \infty$;

(A3) $\mathbb{E}(S^4) < \infty$;

(A4) $\mathbb{E}\left(\log f(Y_i|\theta_i) - \log f(Y_i|\theta^*)\right) > \frac{\beta}{\mathbb{E}(S)}$;

where S is the expected segment length, the expected CPU cost of PELT for analyzing n data points is bounded above by Ln for some constant $L < \infty$.

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